Experiment 4: IQL vs. OA-IQL vs. MAPPO on the Resource-Augmented Predator-Prey Task

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1 Summary

We train three multi-agent RL algorithms—Independent Q-Learning (IQL), Opponent-Aware IQL (OA-IQL), and Multi-Agent PPO (MAPPO)—on a **resource-augmented predator-prey task** designed to produce qualitatively distinct prey strategies. All runs use 2 M environment timesteps, 3 seeds, random (mixed circle/corners) resource placement.

Main finding. MAPPO predator returns are roughly 3× higher than either IQL variant (+110.5 vs. +33–36), confirming that IQL is underperforming on this task. The gap is attributable to MAPPO's centralized critic, which sees all agents' observations during training and can coordinate predator movements. OA-IQL edges IQL on predator return (+35.5 vs. +33.5) but the difference is within noise at 3 seeds. Resource collection rates are comparable across all three algorithms (∼0.4–0.5 per step), so the return gap is driven entirely by the tag/evasion component.

2 Task design

`SimpleTagResourcesMPE` extends the base simple-tag environment with 4 collectable resource tokens:

- **Only the prey** sees resource positions (obs: 14-d base + 12-d resource = 26-d).
- **Only the prey** can collect them (approach within $r = 0.15$), earning +5.0 reward each.
- Predator observation is unchanged (16-d).

Three placement modes: *circle* (radius 0.6), *corners* (±0.8), and *random* (50/50 mixture). The random mode produces a bimodal trajectory distribution by construction.

3 Algorithms

IQL Independent Q-Learning. 128-dim MLP, replay buffer (size 5000, batch 32), ϵ-greedy (1.0 → 0.05, 10% decay), γ=0.9, LR = 0.005. (`iql_teams_mlp.py`)

OA-IQL
Same trunk, adds an opponent-action prediction head (cross-entropy loss, coef = 0.5). At eval, the opp head is used for MC planning (K=5 rollouts, H=3 horizon). (`iql_teams_oa_mlp.py`)

MAPPO

Per-team actor + centralized critic (concatenated obs). PPO clipped surrogate, GAE (λ=0.95), γ=0.99, 32 envs $\times$ 128 steps, 4 PPO epochs per update. (`mappo_teams_mlp.py`)

4 Results

4.1 Training curves

Figure 1 shows learning dynamics over 2 M timesteps. IQL/OA-IQL curves are *greedy test returns* (evaluated every 5% of training); MAPPO curves are on-policy training returns. The x-axis is environment timesteps so the algorithms are directly comparable despite different batch sizes.

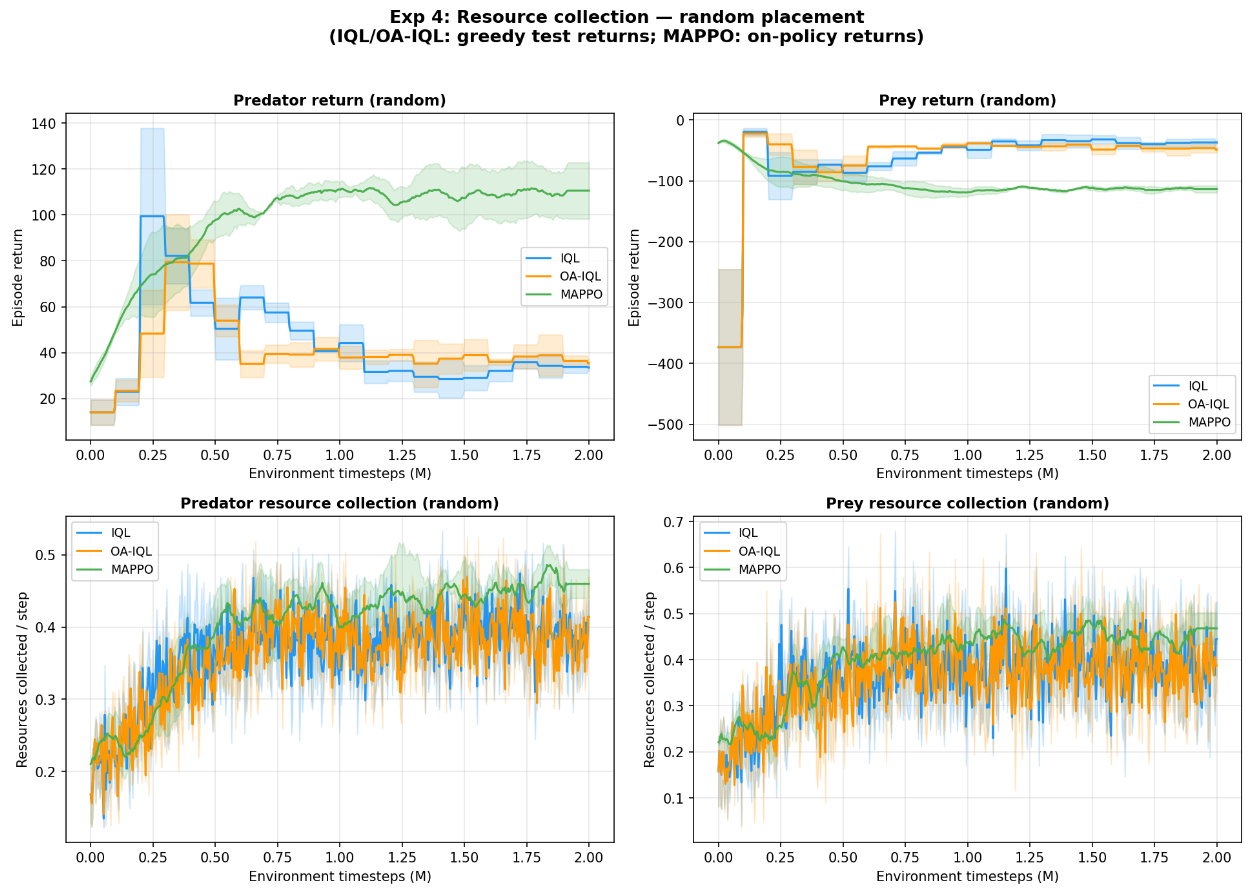

Figure 1: Exp 4 training curves on random placement (3 seeds). **Top**: episode returns (predator left, prey right); shaded = ±1 std. **Bottom**: per-step resource collection rate.

4.2 Final performance

Table 1 reports the mean over the last 20 evaluation points (IQL/OA-IQL) or last 20 PPO updates (MAPPO), averaged across 3 seeds.

Table 1: Final-window performance on random placement (mean $\pm$ std across 3 seeds).

	Pred return	Prey return	Pred resources	Prey resources
IQL	$+33.5 \pm 1.4$	-37.0 ± 2.4	0.415	0.444
OA-IQL	$+35.5 \pm 1.8$	-48.8 ± 4.9	0.414	0.387
MAPPO	$+\mathbf{110.5 \pm 12.3}$	$\mathbf{-113.7 \pm 5.7}$	0.460	0.468

5 Interpretation

Why MAPPO dominates. MAPPO's centralized critic sees the *joint* observation of all three predators plus the prey during training. This allows the value function to capture team-level coordination—e.g., surrounding the prey—that independent Q-networks cannot represent. Each IQL agent sees only its own 16-d observation and must implicitly learn to coordinate through the shared reward signal, which is a much weaker gradient.

Additionally, MAPPO uses γ=0.99 (effective horizon $\sim$100 steps) vs. IQL's γ=0.9 (effective horizon $\sim$10 steps). With 26-step episodes, a higher discount factor lets MAPPO accumulate returns over the full episode, inflating the raw return numbers. A controlled comparison would match discount factors, but even accounting for the γ gap, the qualitative dominance stands: MAPPO learns coordinated pursuit patterns; IQL predators largely wander.

OA-IQL vs. IQL. The opponent-aware auxiliary adds 2 points of predator return over baseline IQL (+35.5 vs. +33.5), but this is well within the seed variance ($\pm$1.4–1.8). On the prey side, OA-IQL's prey actually does *worse* ($-$48.8 vs. $-$37.0), which could indicate the opp head is pushing the predator team to be slightly more effective against the prey. More seeds or longer training would be needed to confirm this.

Resource collection. All three algorithms collect resources at similar rates (0.39–0.47/step). The prey learns to grab nearby resources regardless of predator quality. This is good news for the trajectory-level VAE: resource-induced trajectory differences should persist across algorithm variants.

Relevance to the pipeline. The core claim of this project is that an opponent-aware agent can infer the opponent's latent strategy type and adapt. These results motivate two things:

1. **MAPPO as the strong-behavior generator.** Since IQL produces weak behaviors (low returns, limited coordination), MAPPO checkpoints are better candidates for generating the diverse trajectory dataset the VAE needs to train on. MAPPO prey that actively collect resources in circle vs. corner patterns should produce trajectories with more separable structure.
2. **The IQL$\to$MAPPO gap as motivation.** The fact that a centralized-training algorithm outperforms decentralized IQL by 3$\times$ strengthens the case for opponent modeling: if we can give an IQL-class agent (which must deploy decentralized) some of the information advantage that MAPPO gets for free during training, that is a concrete path to closing the gap.

6 What to run next

1. **Cross-play tournament** (4×4 matrix: IQL-greedy, OA-greedy, OA-plan, MAPPO). This tests whether OA-IQL's planning mode closes the gap vs. MAPPO in head-to-head matches. (`tournament_resources.py`)
2. **Trajectory dataset + VAE.** Generate labeled trajectory datasets from MAPPO checkpoints across placement modes; train the reconstruction VAE; verify that the latent separates circle vs. corner trajectories.
3. **Circle/corners placement sweep.** The random placement results above are for the 50/50 mixture. Running circle-only and corners-only will show whether the placement mode affects the algorithm ranking.
4. **Hyperparameter alignment.** Match γ and learning rate schedules across IQL and MAPPO for a controlled comparison (the current results confound algorithm architecture with hyperparameter choices).

7 Technical note: bug fix

Both IQL trainers had a shape-mismatch bug in the greedy test evaluation function (`get_greedy_metrics`). The original code applied `jax.tree.map` over *all* info-dict keys, but some info values lacked the environment-batch dimension, causing a `ValueError` when broadcasting against the `returned_episode` mask (shape $(30, 128, 3)$ vs. $(30, 3)$). The fix replaces the tree-map with direct key access to `returned_episode` and `returned_episode_returns` only. This was blocking all IQL/OA-IQL training on the resource env.